

Full name: Ha Hoang Phuc Student: 222683 Class: **DH22TIN07 LESSON 1:**

Library management -system - Identify candidate classes (Books, Readers, Loans, Employees, Publishers) + Books: main entity, book information management.

- + DocGia: person who borrows books.
- + PhieuMuon: records borrowing/returning.
- + NhanVien: system manager (librarian, warehouse keeper...).
- + NhaXuatBan: book source information.

- Properties and behavior (methods) of each class +

Book o Attributes: maSach (PK, string), nameSach (string), tacGia (string), namXuatBan (int), maNXB (FK), soLuongTon (int), trangThai (string: "Available"/"On loan").

- o Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(), capNhatTonKho().

+ DocGia o Attributes: maDocGia (PK, string), hoTen (string), daySinh (date), diaChi (string), soDienThoai (string), dayDangKy (date), gioiTinh (boolean).

- o Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon(). +

PhieuMuon o Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK), maNhanVien (FK), nhaMuon (date), ngoiHenTra (date), nhaTraThucTe (date), thanhTrang (string: "On loan"/"Paid"/"Overdue").

- o Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhieuPhat(). + NhanVien o Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), date of birth (date), soDienThoai (string).

- o Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().).

+ NhaXuatBan o Attributes: nameNXB (PK, string), nameNXB (string), address (string), email (string).

- o Behavior: addNXB (), capNhatNXB ().

- Relationships: 2 Books and Publishers: 1

book – 1 publisher

☒☐ Readers and Loans: 1 reader – many loans ☒☐ Books and Loans: 1 book – many loans ☒☐ Employees and Loans: 1 employee – many loans **LESSON 2: Online sales management system** - Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- ☒☐ DangKyTaiKhan (Customer registration).
- ☒☐ DangNhap (Sign in).
- ☒☐ See SanPham (Search / View product list).
- ☒☐ ThemVaoGioHang (Add to cart).
- ☒☐ DatHang (Order).
- ☒☐ ThanhToan (Online payment / COD).
- ☒☐ View DonHang (View order history).

Admin / NguoiQuanLy:

- ☒☐ QuanLySanPham (Add/Edit/Delete products).
- ☒☐ QuanLyDonHang (View/Confirm/Cancel order).
- ☒☐ QuanLyKhachHang (Customer information management).
- ☒☐ QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- ☒☐ ThanhToan includes XacNhanDonHang.
- ☒☐ DatHang extends ThanhToan (if paying online).

- From use case, find necessary classes ☒☐ Customer (Customer): from register, view orders.

- ☒☐ SanPham (Product): product management.
- ☒☐ GioHang (Cart): temporarily store products before ordering.
- ☒☐ Don Hang (Order): from order, payment.
- ☒☐ ChiTietDonHang (OrderDetail): product details in the order.
- ☒☐ ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- ☒☐ Admin or Personnel (inherited from User if there is a user system).

- Analyze the relationship between 2 classes (specify inheritance if 2 any) □ Customer and Customer: 1 customer – many 2 orders (abstract class) ← inherits 2 ← KhachHang and Admin (KhachHang and 2 Admin are both the same type of User, sharing the same account, password, full name).

Lesson 3: Convert class diagrams into data tables

1. MonHoc table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	nameMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maHoHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	rightTao	Date			x	